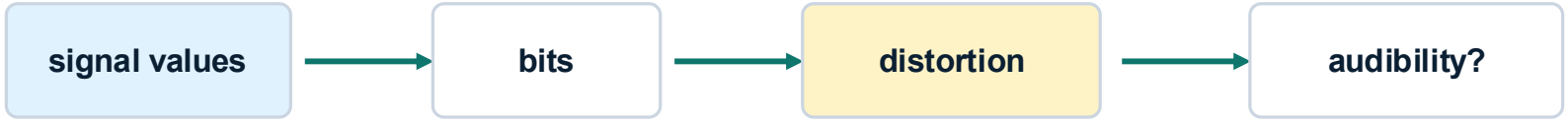
2 Quantization and entropy coding

based on Spanias, Painter & Atti, Chapter 3

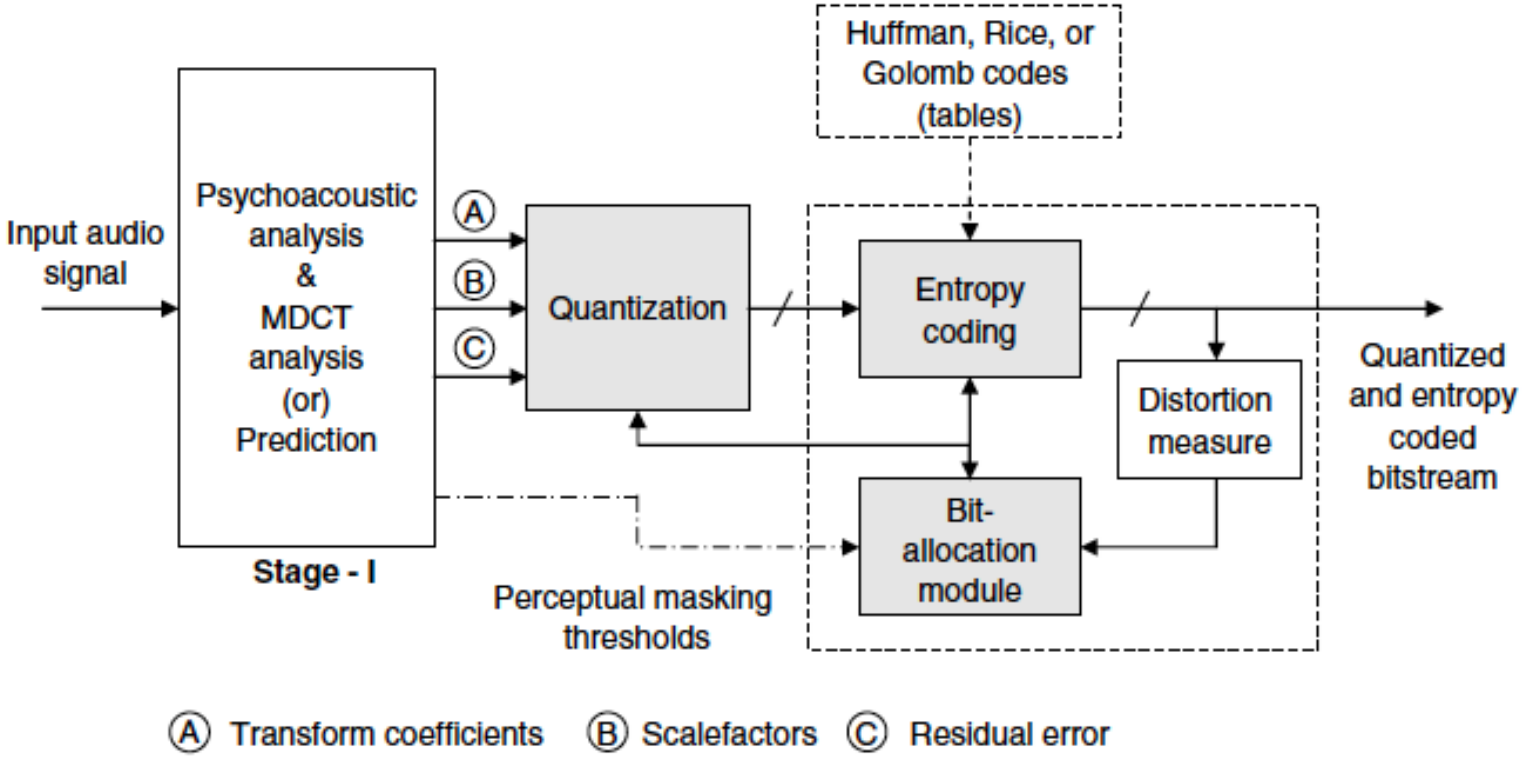


How codecs decide what numerical precision is needed, and how the resulting symbols are represented efficiently.

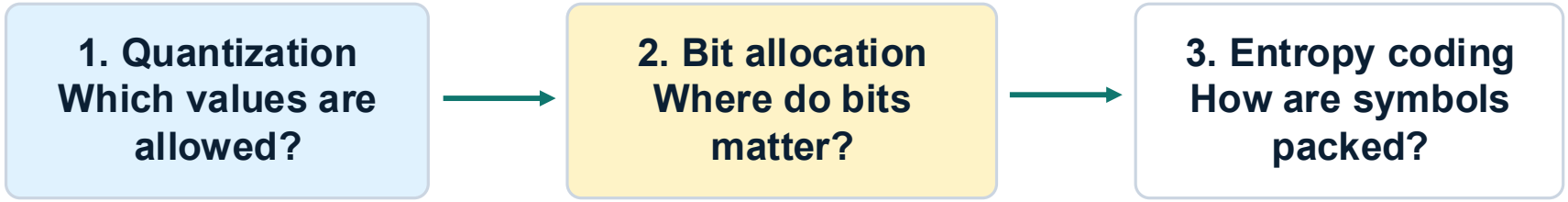
Learning goals

- Explain quantization as controlled approximation of signal values or transform coefficients.
- Relate bit depth, step size, quantization noise and SNR.
- Compare uniform, non-uniform, differential and vector quantization.
- Explain why bit allocation must consider perceptual masking, not only statistical error.
- Construct and interpret a simple Huffman code.

Quantization - Bit Allocation - Entropy coding modules



The three decisions

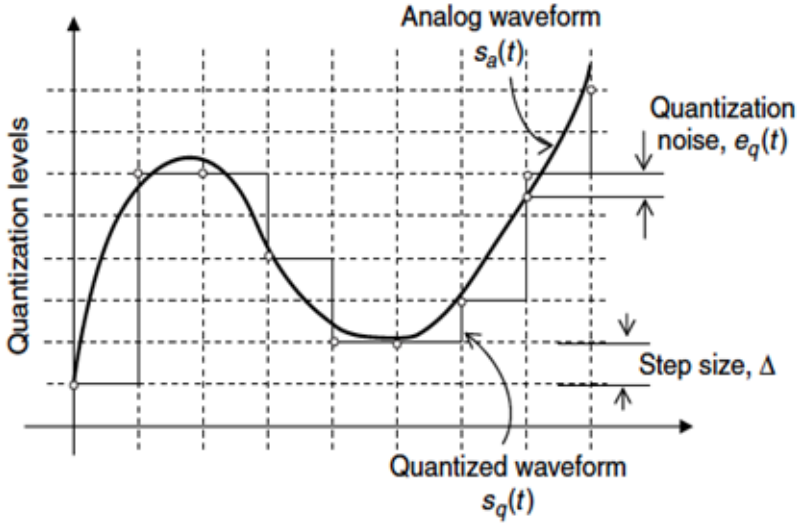


Quantization introduces distortion. Bit allocation shapes that distortion. Entropy coding removes remaining statistical redundancy without adding distortion.

Quantization: the basic idea

Replace a continuum of possible amplitudes by a finite set of reconstruction levels.

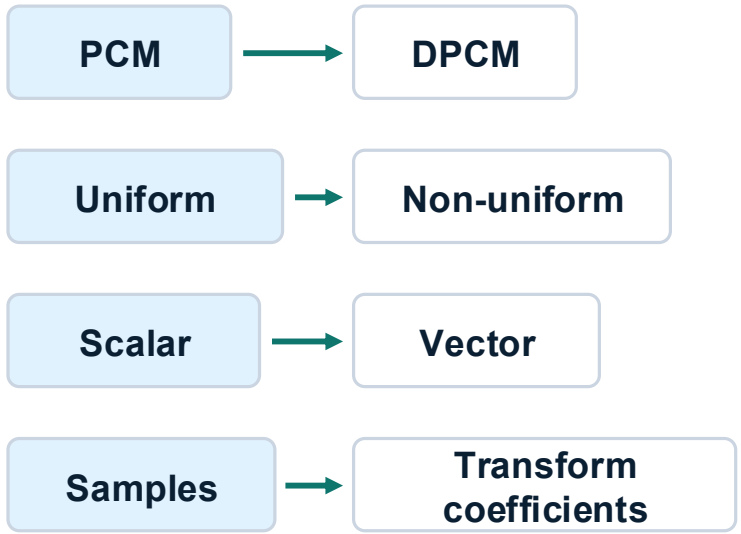
- The input sample or coefficient is mapped to a nearby allowed value.
- The difference is quantization error: e[n] = x[n] − x̂[n].
- In audio coding, this error is not only minimized; it is shaped to be less audible.



A taxonomy of quantizers

Different quantizers exploit different kinds of structure.

- Memoryless vs. predictive: are past samples used?
- Uniform vs. non-uniform: are all step sizes equal?
- Scalar vs. vector: are values coded individually or in blocks?
- Waveform vs. parametric/transform: what representation is quantized?



Probability density functions

A random process can be characterized by its probability density function (PDF), which is a non-negative function p(x), whose properties are

$$\int_{-\infty}^{\infty} p(x)dx = 1$$

and

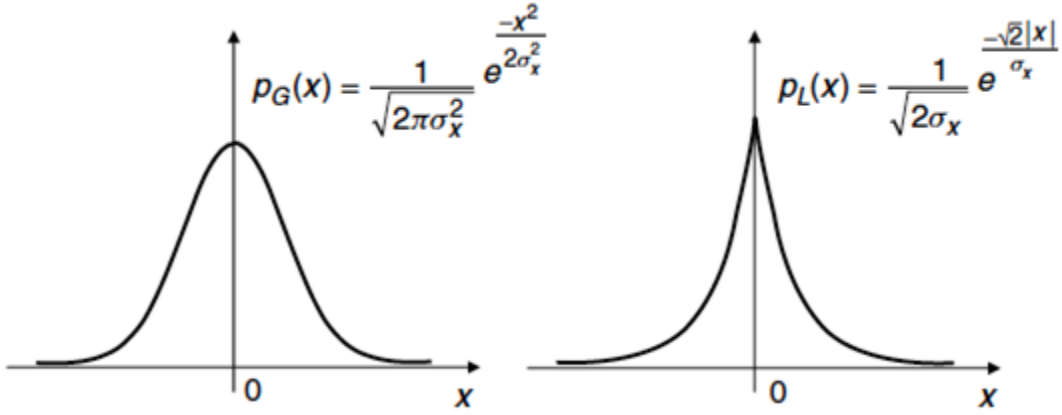
$$\int_{x_1}^{x_2} p(x)dx = \Pr(x_1 < X \leqslant x_2).$$

the PDF area, from x1 to x2, is the probability that the random variable X is observed in this range.

The mean and the variance of X are defined as

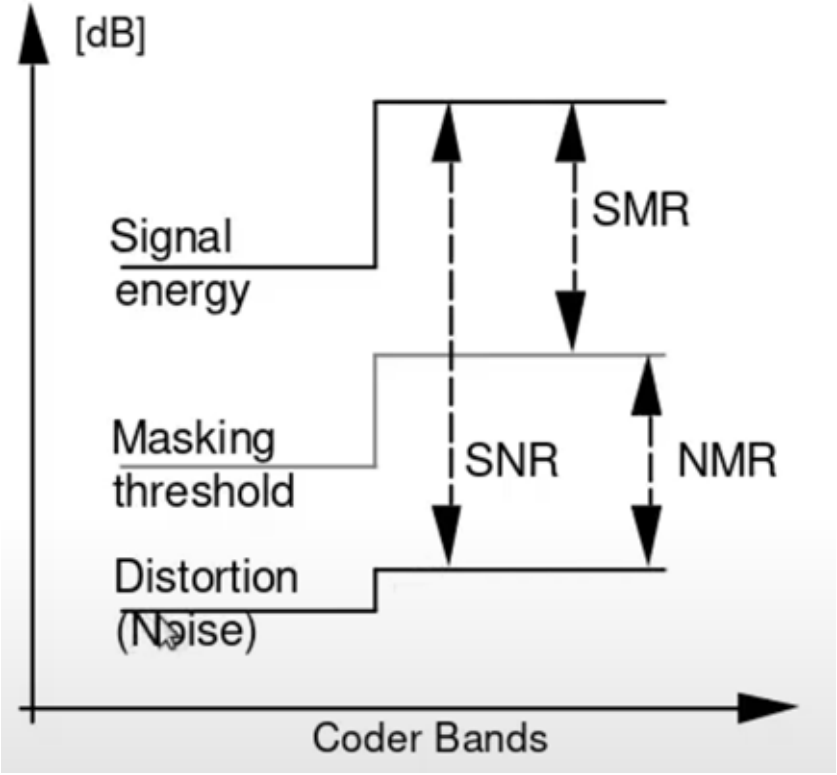
$$\mu_x = E[X] = \int_{-\infty}^{\infty} x p(x) dx \qquad \sigma_x^2 = \int_{-\infty}^{\infty} (x - \mu_x)^2 p(x) dx = E[(X - \mu_x)^2]$$

PDFs are useful in the design of optimal signal quantizers as they can be used to determine the assignment of optimal quantization levels.



Gaussian PDF Laplacian PDF

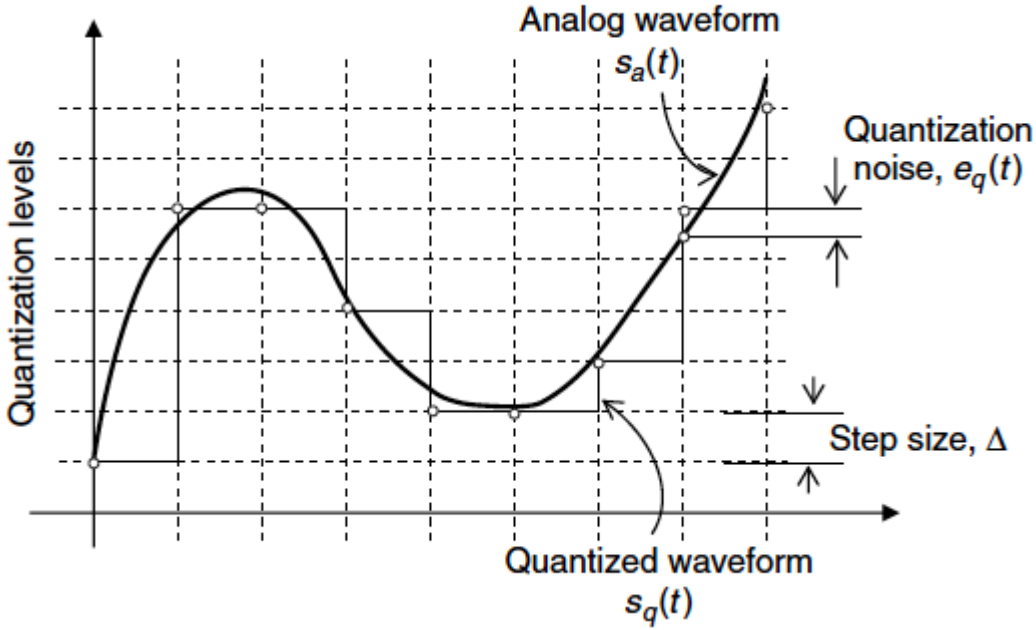
Quality and rate measures

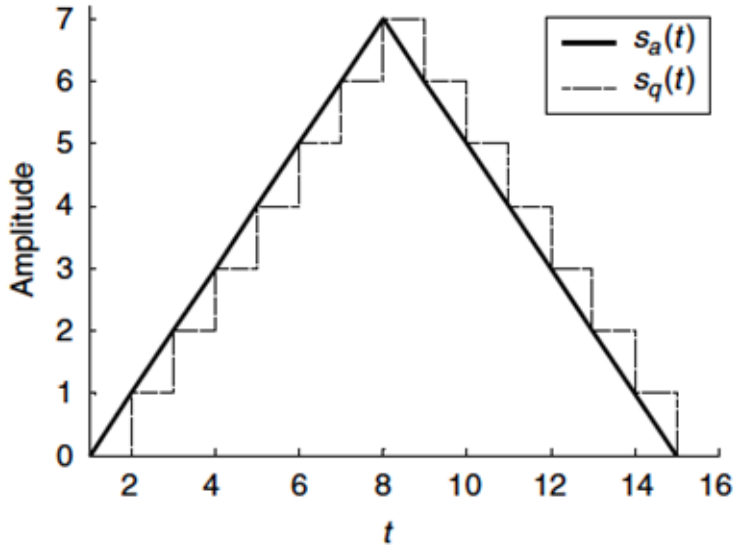


- Signal-to-Noise Ratio (SNR)
  - Quadratic distortion measure
- Noise-to-Mask Ratio (NMR)
  - Ratio of distortion with respect to masking threshold
  - Determines audibility of distortion
  - Should be as small as possible (<= 0 dB)
- Signal-to-Mask Ratio (SMR)
  - Relation between signal and masking threshold
  - Gives indication of bit demand

Scalar quantization: uniform

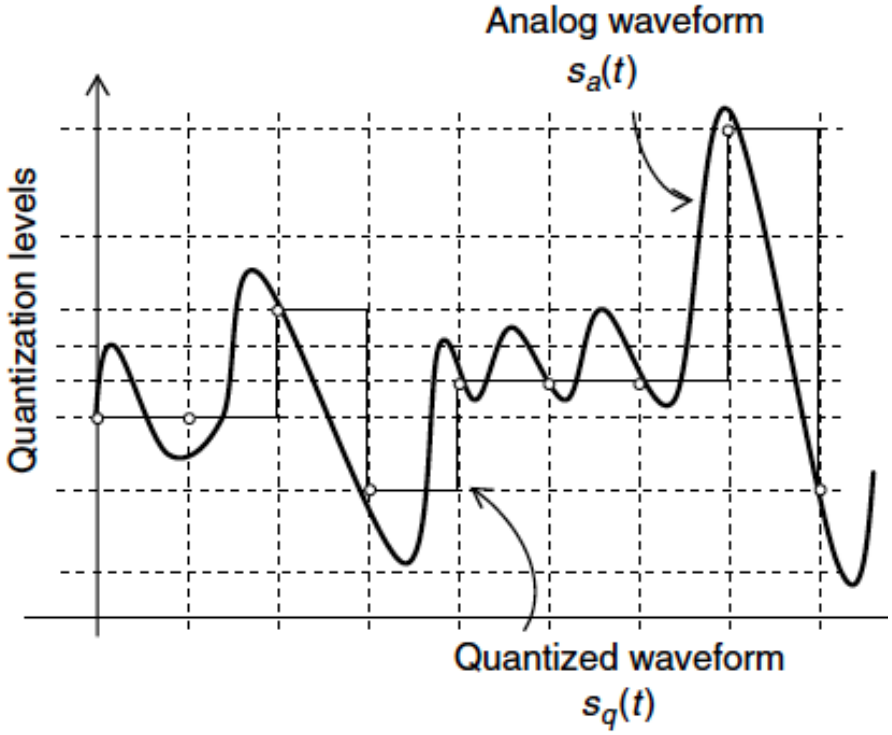
Uniform PCM is a memoryless process that quantizes amplitudes by rounding off each sample to one of a set of discrete values.

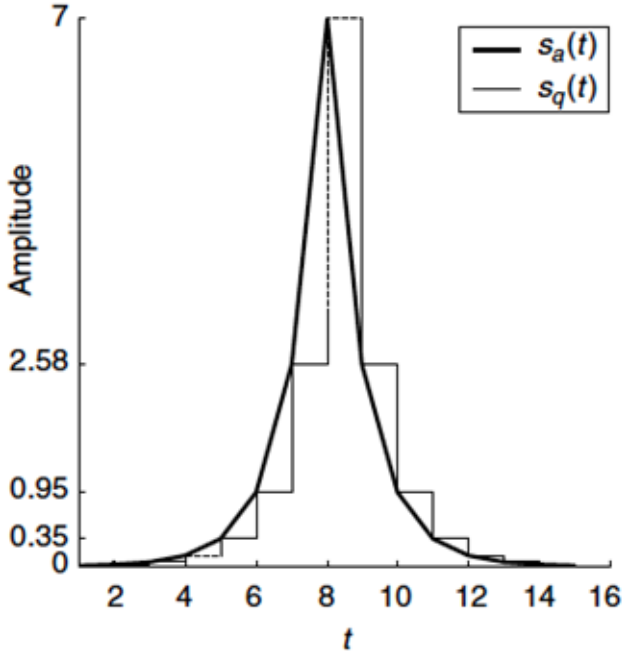




- The difference between adjacent quantization levels, i.e., the step size Δ, is constant.
- The number of quantization levels, Q, is Q = 2^Rb, where Rb denotes the number of bits.
- The performance of uniform PCM can be described in terms of signal-to-noise ratio, SNR = $10\log_{10}\left(\frac{\sum_n x^2(n)}{\sum_n (x(n)-\hat{x}(n))^2}\right)(\text{dB})$
- Consider a signal s is quantized in the interval (-smax, smax). A uniform step size is $\Delta = \frac{2s_{\max}}{2^{R_b}}$.
- Let's assume a quantization noise with uniform PDF, $p_{e_q}(e_q) = \frac{1}{\Delta}$, for $|e_q| \leqslant \frac{\Delta}{2}$.
- The variance of the quantization noise is $\sigma_{e_q}^2 = \frac{\Delta^2}{12} = \frac{s_{\max}^2 2^{-2R_b}}{3}$.
- Increase of 1 bit reduces the noise variance by a factor of four and improves the SNR approximately by 6dB.

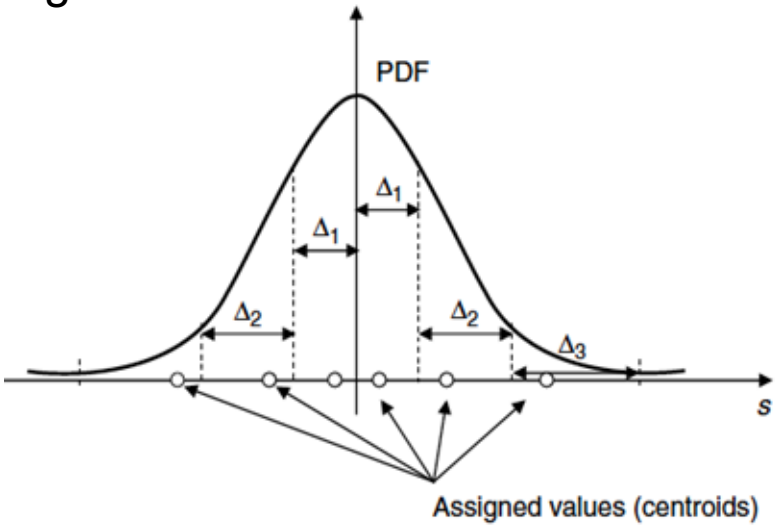
Scalar quantization: non uniform



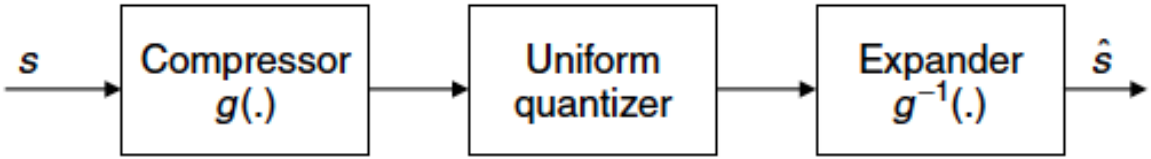


PDF-optimized PCM

- Non-uniform PCM quantizers use a non uniform step size that can be determined from the statistical structure of the signal.
- PDF-optimized PCM uses fine step sizes for frequently occurring amplitudes and coarse step sizes for less frequently occurring amplitudes.
- A signal with a Gaussian PDF can be quantized more efficiently by computing the quantization step sizes and the corresponding centroids such that the mean square quantization noise is minimized.



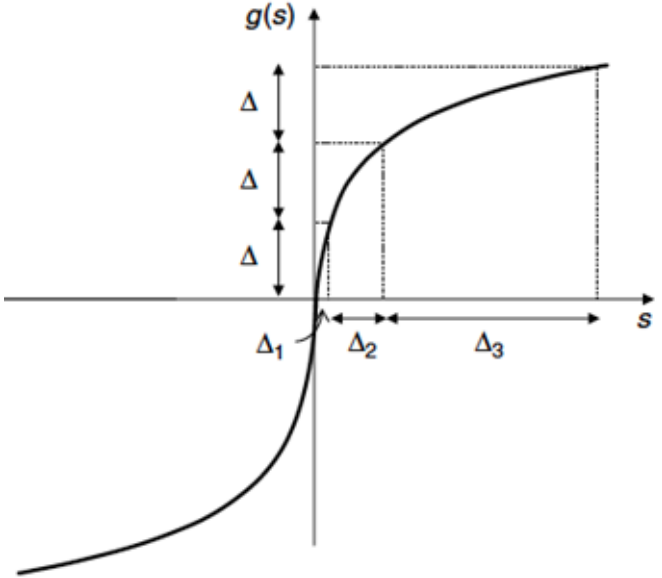
Log quantizers



g(.) maps non-uniform step sizes to uniform step sizes, such that a simple linear quantizer is used.

Two telephony standards have been developed based on logarithmic companding, i.e., the μ-law (in USA) and the A-law (in Europe).

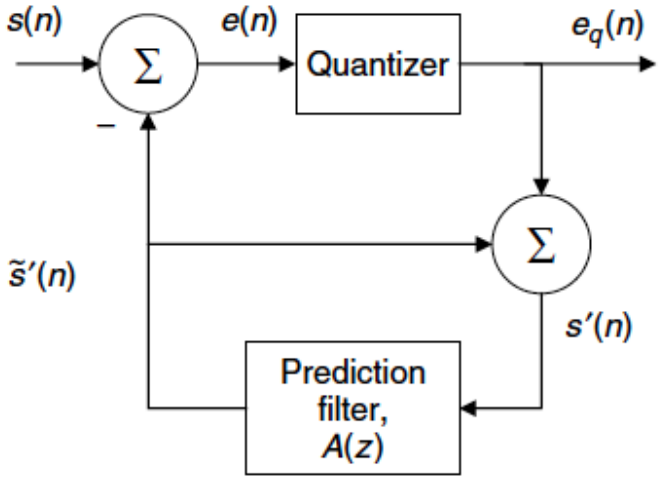
Companding schemes have been found to reduce bit rates, without degradation, by as much as 4 bits/sample relative to uniform PCM.

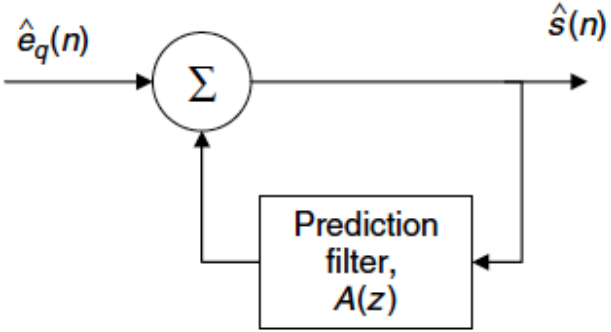


Companding function for non uniform PCM

Differential PCM

- DPCM removes redundancy in the signal by exploiting the correlation between adjacent samples.
- Simplest DPCM encodes only the difference between successive samples.
- Common DPCM incorporates short-term prediction process: $A(z) = \sum_{i=1}^{p} a_i z^{-i}$



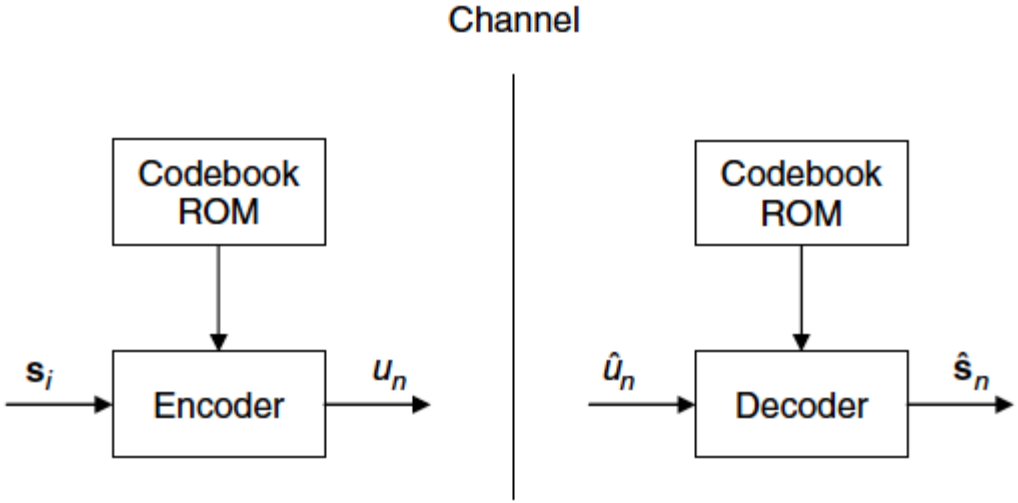


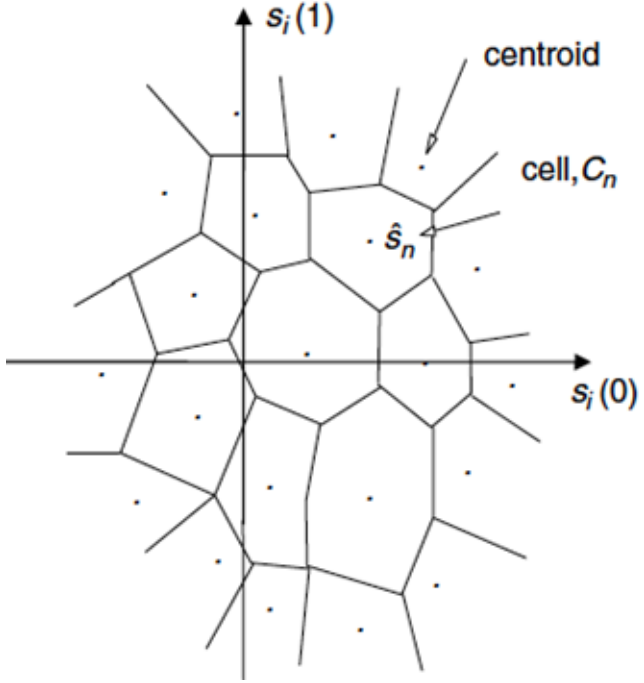
Common PCM formats based on scalar quantization

PCM-Format	DVD-Audio (typ.)	DAT (typ.)	CD	Speech Communication (7kHz)	Speech Communication ITU-T G.711
Sampling Frequency	96 kHz	48 kHz	44,1 kHz	16 kHz	8 kHz
Quantization	24 bit **uniform**	16 bit **uniform**	16 bit **uniform**	16 bit **uniform**	8 bit **Non uniform**
Num chanels	2	2	2	1	1
Bitrate	2 x 2,3 Mbit/s	2 x 768 kbit/s	2 x 706 kbit/s	256 kbit/s	64 kbit/s

Vector quantization

VQ is achieved by encoding a data-set jointly in block or vector form.





Cells for two-dimensional VQ

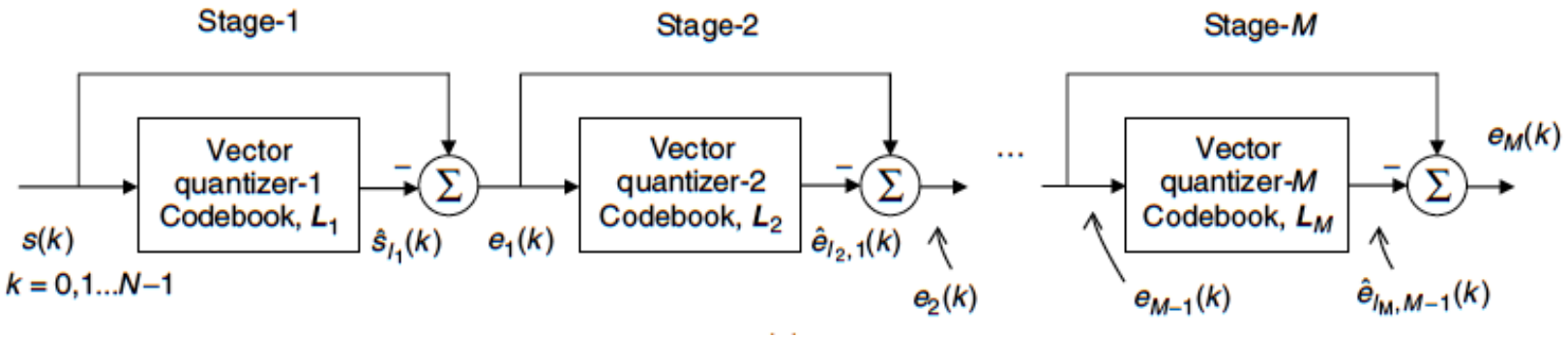
- The quantizer maps the i-th incoming N vector given by $\mathbf{s}_i = [s_i(0), s_i(1), \ldots, s_i(N-1)]^T$ to a n-th channel symbol u_n, $n = 1,|2, \ldots, L$
- The codebook consists of L code vectors, $\hat{\mathbf{s}}_n = [\hat{s}_n(0), \hat{s}_n(1), \ldots, \hat{s}_n(N-1)]^T, n = 1, 2, .., L,$ which reside in the memory of the transmitter and the receiver.
- The input vectors are compared to each codeword and the address of the closest codeword determines the channel symbol to be transmitted.
- The most commonly used distortion measure is the sum of squared errors,

$$\varepsilon(\mathbf{s}_i, \hat{\mathbf{s}}_n) = \sum_{k=0}^{N-1} (s_i(k) - \hat{s}_n(k))^2$$

- Fixed codebooks are designed a priori and the basic design procedure involves an initial guess for the codebook and then iterative improvement by using a large number of training vectors.

Structured VQ

- The complexity of high-dimensionality VQ can be reduced with the use of structured codebooks that allow for efficient search.
- Multistep vector quantizers consist of a cascade of several quantizers each one encoding the error or residual of the previous quantizer.



Bit-allocation algorithms

- A bit-allocation algorithm determines the number of bits required to quantize an audio frame with reduced audible distortion.
- Bit-allocation can be based on perceptual rules or spectral characteristics.

Table 3.1. Uniform bit-allocation scheme, where $n_i = [N/N_f]$, $\forall i \in [1, N_f]$.

Uniformly distributed coefficients		Gaussian-distributed coefficients	
Input vector, $\mathbf{x}_u$	Quantized vector, $\hat{\mathbf{x}}_u$	Input vector, $\mathbf{x}_g$	Quantized vector, $\hat{\mathbf{x}}_g$
[0.6029, 0.3806, 0.56222, 0.12649, 0.26904, 0.47535, 0.4553, 0.38398, 0.41811, 0.35213, 0.23434, 0.32256, 0.31352, 0.3026, 0.32179, 0.16496]	[0.625, 0.375, 0.5625, 0.125, 0.25, 0.5, 0.4375, 0.375, 0.4375, 0.375, 0.25, 0.3125, 0.3125, 0.3125, 0.3125, 0.1875]	[0.5199, 2.4205, −0.94578, −0.0081113, −0.42986, −0.87688, 1.1553, −0.82724, −1.345, −0.15859, −0.23544, 0.85353, 0.016574, −2.0292, 1.2702, 0.28333]	[0.5, 2.4375, −0.9375, 0, −0.4375, −0.875, 1.125, −0.8125, −1.375, −0.1875, −0.25, 0.875, 0, −2, 1.25, 0.3125]

$D = \frac{1}{N_f}\sum_{i=1}^{N_f} \mathrm{E}[(x_i - \hat{x}_i)^2]$, $n_i = 4 \forall i \in [1, 16]$, $D_u = 0.00023927$ and $D_g = 0.00042573$

N = 64 bits available, N_f = 16 samples -> n_i = 4 for all i

We can define a cost function that minimizes the distortion

$$\min_{n_i}\{D\} = \min_{n_i}\left\{\frac{1}{N_f}\sum_{i=1}^{N_f}\mathrm{E}[(x_i - \hat{x}_i)^2]\right\} = \min_{n_i}\left\{\frac{1}{N_f}\sum_{i=1}^{N_f}\sigma_i^2\right\}$$

Table 3.2. Optimal bit-allocation scheme, where $n_i^{optimum} = \frac{N}{N_f} + \frac{1}{2}\log_2 x_i^2 - \frac{1}{2}\log_2\left(\prod_{i=1}^{N_f} x_i^2\right)^{\frac{1}{N_f}}$.

Uniformly distributed coefficients		Gaussian-distributed coefficients	
Input vector, $\mathbf{x}_u$	Quantized vector, $\hat{\mathbf{x}}_u$	Input vector, $\mathbf{x}_g$	Quantized vector, $\hat{\mathbf{x}}_g$
[0.6029, 0.3806, 0.56222, 0.12649, 0.26904, 0.47535, 0.4553, 0.38398, 0.41811, 0.35213, 0.23434, 0.32256, 0.31352, 0.3026, 0.32179, 0.16496]	[0.59375, 0.375, 0.5625, 0.125, 0.25, 0.46875, 0.4375, 0.375, 0.4375, 0.375, 0.25, 0.3125, 0.3125, 0.3125, 0.3125, 0.125]	[0.5199, 2.4205, −0.94578, −0.0081113, −0.42986, −0.87688, 1.1553, −0.82724, −1.345, −0.15859, −0.23544, 0.85353, 0.016574, −2.0292, 1.2702, 0.28333]	[0.5, 2.4219, −0.9375, 0, −0.4375, −0.875, 1.1563, −0.8125, −1.3438, −0.125, −0.25, 0.84375, 0, −2.0313, 1.2656, 0.25]
Bits allocated, n = [5 4 5 3 4 5 4 4 4 4 4 4 4 4 4 3] Distortion, $D_u = 0.0002468$			
Bits allocated, n = [4 6 5 −2 4 5 5 5 6 3 3 5 −1 6 6 3] Distortion, $D_g = 0.00022878$			

Entropy coding

Definition: lossless compression based on exploiting statistical correlations of the data.

- Entropy is a measure of uncertainty of a random variable.
- Shannon proved that the minimum number of bits required to encode a message, X, is given by the entropy, $H_e(X) = -\sum_{i=1}^{K} p_i \log_2(p_i)$, where p_i is the probability that the i-th symbol is transmitted.
- Typical probability of audio samples: Laplace distribution.

Swimmer	Probability of winning	Binary string or the identifier
S_1	1/2	0
S_2	1/4	10
S_3	1/8	110
S_4	1/16	1110
S_5	1/64	111100
S_6	1/64	111101
S_7	1/64	111110
S_8	1/64	111111

$$H_e(X) = -\sum_{i=1}^{8} p_i \log_2(p_i)$$
$$= -\left\{\frac{1}{2}\log_2\left(\frac{1}{2}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{1}{8}\log_2\left(\frac{1}{8}\right) + \frac{1}{16}\log_2\left(\frac{1}{16}\right) + \frac{4}{64}\log_2\left(\frac{1}{64}\right)\right\}$$
$$= 2.$$

-> 2 bits with entropy code, 3 bits with uniform code.

Example of an entropy code

Input bitstream, [4 5 6 6 2 5 4 4 1 4 4]

(*a*) *Encoding based on the binary equal - length code* :

[100 101 110 110 010 101 100 100 001 100 100]

Total length, $L_b = 33$ *bits*

(*b*) *Encoding based on the Shannon - Fano Coding* :

4'*s-5 times*, 5'*s-2 times*, 6'*s-2 twice*, 2'*s-once*, *and* 1'*s-once*

Probabilities, $p_4 = 5/11$, $p_5 = 2/11$, $p_6 = 2/11$, $p_2 = 1/11$, *and* $p_1 = 1/11$

Representations : $4 \rightarrow 0$, $5 \rightarrow 10$, $6 \rightarrow 110$, $1 \rightarrow 1110$, $2 \rightarrow 11110$

[0 10 110 110 11110 10 0 0 1110 0 0]

Total length, $L_{SF} = 24$ *bits*

Figure 3.16. Entropy coding schemes: (a) binary equal-length code, (b) Shannon-Fano coding.

- Statistical entropy alone does not provide a good measure of compressibility in the case of audio coding. Quantization noise, masking thresholds, and tone- and noise masking effects must be accounted for.
- Most popular entropy coding approach: Huffman coding.

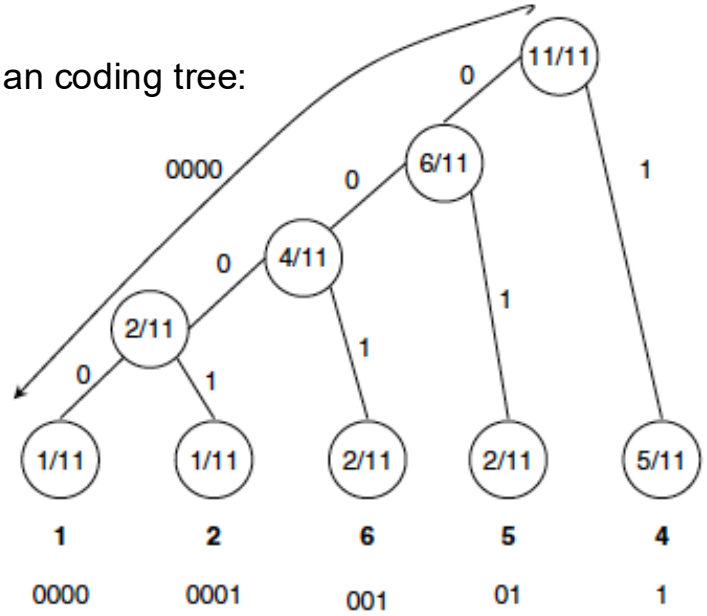
Huffman coding

- Technique to construct minimum redundancy codes.
- Uses the probability of the symbol to determine the codeword.
- Regarded as the most effective compression method, provided that the codes designed using a specific set of symbol frequencies match the input symbol frequencies.

Table 3.4. Huffman codetable for the input bitstream, $X = [4\ 5\ 6\ 6\ 2\ 5\ 4\ 4\ 1\ 4\ 4]$.

Input symbol	Probability	Huffman codeword
4	5/11	1
5	2/11	01
6	2/11	001
2	1/11	0001
1	1/11	0000

Huffman coding tree:



Entropy of the input bitstream X:

$$H_e(X) = -\sum_{i=1}^{K} p_i \log_2(p_i)$$

$$= -\left\{\frac{2}{11}\log_2\left(\frac{1}{11}\right) + \frac{4}{11}\log_2\left(\frac{2}{11}\right) + \frac{5}{11}\log_2\left(\frac{5}{11}\right)\right\}$$

$$= 2.04.$$

Total length of the encoded bitstream is L_HF = 23 bits.

Implementation

1. **Calculate Frequencies:** calculate the frequency of each character in the input array.
2. **Build the Huffman Tree:** the tree is built by merging the two least frequent nodes until there is only one node left. This node becomes the root of the tree.
3. **Generate Huffman Codes:** the codes are generated by traversing the tree. A left edge corresponds to appending '0' to the code, and a right edge corresponds to appending '1'.
4. **Encode the Input:** the input array is encoded using the generated codes.
5. **Decode the Encoded Data:** the encoded data is decoded by traversing the tree.

Notebook examples for class

- examples/downsample-chirp.ipynb — revisit sampling and aliasing before quantization.
- examples/resample.ipynb — practical rate conversion and discrete-time representations.
- examples/entropy.ipynb — estimate entropy from symbol counts and probabilities.
- examples/huffman-coding.ipynb — build, encode and decode a Huffman code.

Repository: github.com/xserra/audio-coding-materials

What to listen for in demos

- Coarse quantization: granular noise, distortion on low-level signals.
- Insufficient bits in a critical band: noise becomes audible above masking threshold.
- Predictive coding: smaller residual when the signal is smooth or correlated.
- Entropy coding: same reconstruction, fewer bits.

Key takeaways

- Quantization is the lossy step: it trades precision for bit rate.
- The distribution of values determines efficient quantization levels.
- Bit allocation decides where precision is perceptually valuable.
- Entropy coding is lossless and exploits unequal symbol probabilities.
- Modern perceptual audio codecs combine all these ideas in each frame.